DAA EXPERIMENT 5

Out of Boundary Paths

Aim

To calculate the number of paths that move outside the boundary of a grid within a given number of moves.

Algorithm

1. Start from the given cell in the grid.
2. If the current position is outside the grid, return 1.
3. If no moves remain, return 0.
4. Otherwise, recursively explore the four directions: up, down, left, and right.
5. Add the results from all four directions and return the total.
6. Display the number of paths.
7. Stop.

Program

```
# Out of Boundary Paths - Recursive Approach

def paths(m, n, moves, i, j):
    if i < 0 or i >= m or j < 0 or j >= n:
        return 1

    if moves == 0:
        return 0

    return (paths(m, n, moves - 1, i + 1, j) +
            paths(m, n, moves - 1, i - 1, j) +
            paths(m, n, moves - 1, i, j + 1) +
            paths(m, n, moves - 1, i, j - 1))


m = int(input("Enter number of rows: "))
n = int(input("Enter number of columns: "))
moves = int(input("Enter maximum moves: "))
i = int(input("Enter starting row: "))
j = int(input("Enter starting column: "))

print("Number of paths:", paths(m, n, moves, i, j))
```

Complexity Analysis

- Time Complexity: O(4^N), where N is the maximum number of moves
- Space Complexity: O(N) recursion stack

Output

```
Enter number of rows: 2
Enter number of columns: 2
Enter maximum moves: 2
Enter starting row: 0
Enter starting column: 0
Number of paths: 6
PS C:\Users\jithe\OneDrive\Documents\DAA - programs>
```

Result

Thus, the number of paths that move out of the grid boundary was calculated using recursion.